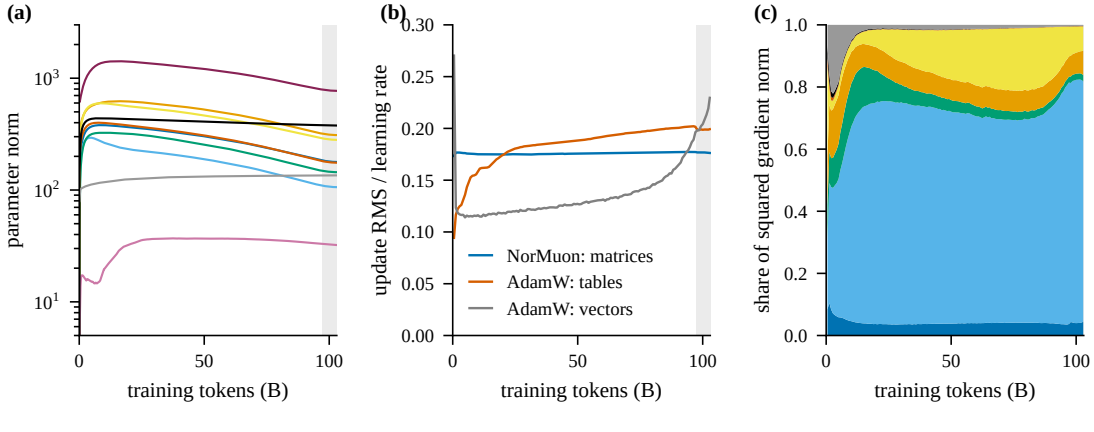
(a)
parameter norm
10^3
10^2
10^1
0
50
100
training tokens (B)
(b)
update RMS / learning rate
0.30
0.25
0.20
0.15
0.10
0.05
0.00
NorMuon: matrices
AdamW: tables
AdamW: vectors
0
50
100
training tokens (B)
(c)
share of squared gradient norm
1.0
0.8
0.6
0.4
0.2
0.0
0
50
100
training tokens (B)
attention q, k
attention v
attention output
MLP input
MLP output
value-embedding gates
value-embedding tables
token embedding
output layer
vectors